

Сетчатка

Сетчатка (от лат. *rete* 'сеть'; мн. ч. *retinae* или *retinas*) — это внутренний, светочувствительный слой **ткани глаза** большинства **позвоночных** и некоторых **моллюсков**. Оптическая система глаза создает на сетчатке **сфокусированное** двухмерное изображение окружающего мира, которое затем обрабатывается сетчаткой и посылает нервные импульсы по **зрительному нерву** в **зрительную кору** для формирования **зрительного восприятия**. Сетчатка выполняет функцию, во многом аналогичную функции **пленки** или **матрицы** в **камере**.

Нервная сетчатка состоит из нескольких слоев **нейронов**, соединенных между собой **синапсами**, и поддерживается внешним слоем пигментированных эпителиальных клеток. Основными светочувствительными клетками сетчатки являются **фоторецепторы**, которые бывают двух типов: **палочки** и **колбочки**. Палочки функционируют в основном при тусклом освещении и обеспечивают монохромное зрение. Колбочки функционируют при хорошем освещении и отвечают за восприятие цвета с помощью ряда **опсинов**, а также за остроту зрения, необходимую для таких задач, как чтение. Третий тип светочувствительных клеток, **светочувствительные ганглиозные клетки**, важен для **синхронизации циркадных ритмов** и рефлекторных реакций, таких как **зрачковый рефлекс**.

Свет, попадающий на сетчатку глаза, запускает каскад химических и электрических процессов, которые в конечном итоге приводят к возникновению нервных импульсов, передаваемых в различные зрительные центры **мозга** по волокнам **зрительного нерва**. Нервные сигналы от палочек и колбочек обрабатываются другими нейронами, которые генерируют **потенциалы действия** в **ганглиозных клетках сетчатки**, **аксоны** которых образуют зрительный нерв. ^[1]

У позвоночных в **процессе эмбрионального развития** сетчатка и зрительный нерв образуются как выросты развивающегося головного мозга, в частности эмбрионального **диэнцефалона**. Таким образом, сетчатка считается частью **центральной нервной системы** (ЦНС) и фактически является тканью головного мозга. ^{[2][3]} Это единственная часть ЦНС, которую можно визуализировать **неинвазивно**. Как и большая часть мозга, сетчатка изолирована от сосудистой системы **гематоэнцефалическим барьером**. Сетчатка глаза — это часть тела, которая больше всего нуждается в постоянной подпитке энергией. ^[4]

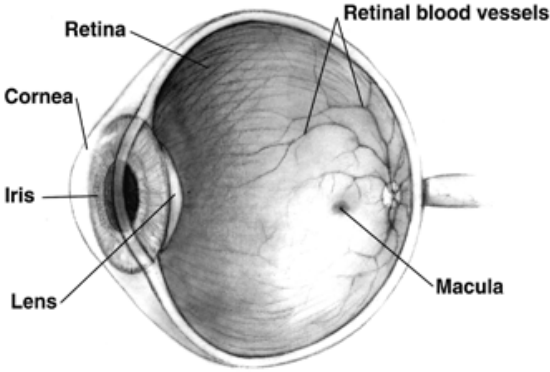
Структура

Перевернутая и обычная сетчатка

Сетчатка глаза позвоночных *инвертирована* в том смысле, что светочувствительные клетки расположены в задней части сетчатки, поэтому свет должен пройти через слои нейронов и капилляров, прежде чем достигнет светочувствительных участков палочек и колбочек. [5] Ганглиозные клетки, аксоны которых образуют зрительный нерв, расположены в передней части сетчатки, поэтому зрительный нерв должен пройти через сетчатку, чтобы попасть в мозг. В этой области нет фоторецепторов, что приводит к образованию **слепого пятна**. [6] В сетчатке **головоногих** моллюсков фоторецепторы расположены спереди, а за ними находятся обрабатывающие нейроны и капилляры. Поэтому у головоногих моллюсков нет слепого пятна.

Although the overlying neural tissue is partly transparent, and the accompanying **glial cells** have been shown to act as **fibre-optic** channels to transport photons directly to the photoreceptors, [7][8] **light scattering** does occur. [9] Some vertebrates, including humans, have an area of the central retina adapted for high-acuity vision. This area, termed the **fovea centralis**, is avascular (does not have blood vessels), and has minimal neural tissue in front of the photoreceptors, thereby minimizing light scattering. [9]

The cephalopods have a non-inverted retina, which is comparable in **resolving power** to the eyes of many vertebrates. Squid eyes do not

Сетчатка	
	
Правый человеческий глаз в разрезе. Глаза у разных животных устроены по-разному.	
Подробные сведения	
Произношение	/'retinə/ Великобритания: , /'retənə/США: , мн. ч. сетчатки /-ni/
Часть	Глаз
Система	Визуальная система
Артерия	Центральная артерия сетчатки
Идентификаторы	
Латинский	<i>rēte</i> , внутренняя оболочка луковицы
Сетка	D012160 (https://meshb.nlm.nih.gov/record/ui?ui=D012160)
TA98	A15.2.04.002 (https://ifaa.unifr.ch/Public/EntryPage/TA98%20Tree/Entity%20TA98%20EN/15.2.04.002%20Entity%20TA98%20EN.htm)

have an analog of the vertebrate [retinal pigment epithelium](#) (RPE). Although their photoreceptors contain a protein, retinochrome, that recycles retinal and replicates one of the functions of the vertebrate RPE, cephalopod photoreceptors are likely not maintained as well as in vertebrates, and that as a result, the useful lifetime of photoreceptors in invertebrates is much shorter than in vertebrates.^[10] Having easily replaced stalk eyes (some lobsters) or retinae (some spiders, such as *Deinopis*^[11]) rarely occurs.

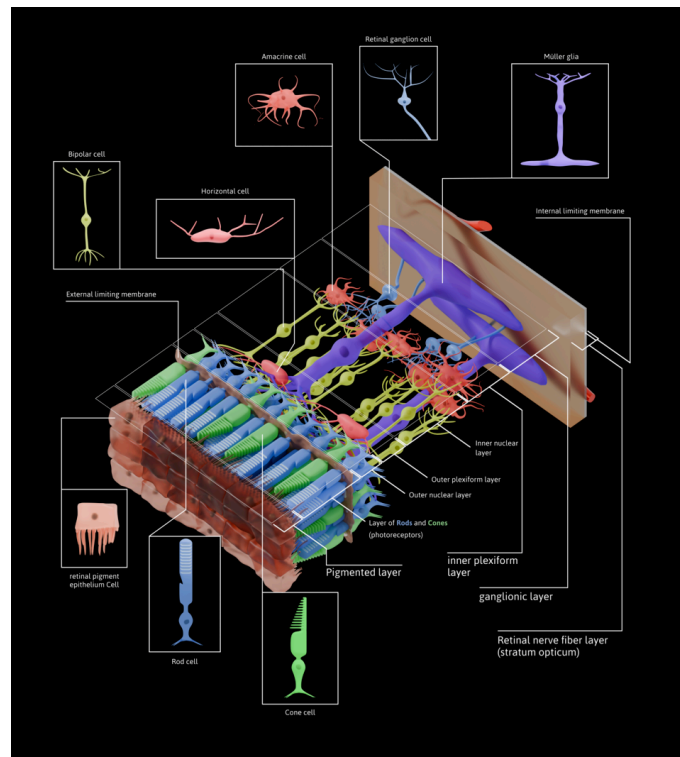
TA2	6776 (https://ta2viewer.openanatomy.org/?id=6776)
FMA	58301 (https://bioportal.bioontology.org/ontologies/FMA/?p=classes&conceptid=http%3A%2F%2Fpurl.org%2Fsig%2Font%2Ffma%2Ffma58301)
Анатомическая терминология	

The cephalopod retina does not originate as an outgrowth of the brain, as the vertebrate one does. This difference suggests that vertebrate and cephalopod eyes are not [homologous](#), but have evolved separately. From an evolutionary perspective, a more complex structure such as the inverted retina can generally come about as a consequence of two alternate processes - an advantageous "good" compromise between competing functional limitations, or as a historical maladaptive relic of the convoluted path of organ evolution and transformation. Vision is an important adaptation in higher vertebrates.

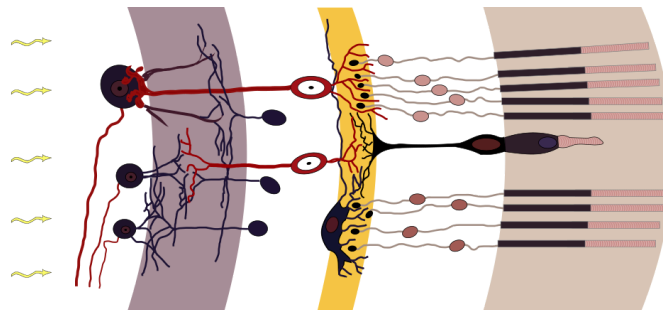
A third view of the "inverted" vertebrate eye is that it combines two benefits - the maintenance of the photoreceptors mentioned above, and the reduction in light intensity necessary to avoid blinding the photoreceptors, which are based on the extremely sensitive eyes of the ancestors of modern hagfish (fish that live in very deep, dark water).^[12]

An APS (American Physical Society) study on the evolutionary purpose for the inverted retina structure^[13] says that "The directional of glial cells helps increase the clarity of human vision. But we also noticed something rather curious: the colours that best passed through the glial cells were green to red, which the eye needs most for daytime vision. The eye usually receives too much blue—and thus has fewer blue-sensitive cones. Further computer simulations showed that green and red are concentrated five to ten times more by the glial cells, and into their respective cones, than blue light. Instead, excess blue light gets scattered to the surrounding rods. This optimization is such that color vision during the day is enhanced, while night-time vision suffers very little".^[13]

Retinal layers



an orthographic cross-section of the layers of the human retina labeling various elements. Light comes from top right



Rods, cones, and nerve layers in the retina: The front (anterior) of the eye is on the left. Light (from the left) passes through several transparent nerve layers to reach the rods and cones (far right). Chemical changes in the rods and cones send a signal back to the nerves. The signal goes first to the **bipolar** and **horizontal cells** (yellow layer), then to the **amacrine cells** and **ganglion cells** (purple layer), then to the optic nerve fibres. The signals are processed in these layers. First, the signals start as raw outputs of points in the rod and cone cells. Then, the nerve layers identify simple shapes, such as bright points surrounded by dark points, edges, and movement. (Based on a drawing by [Ramón y Cajal, 1911](#))

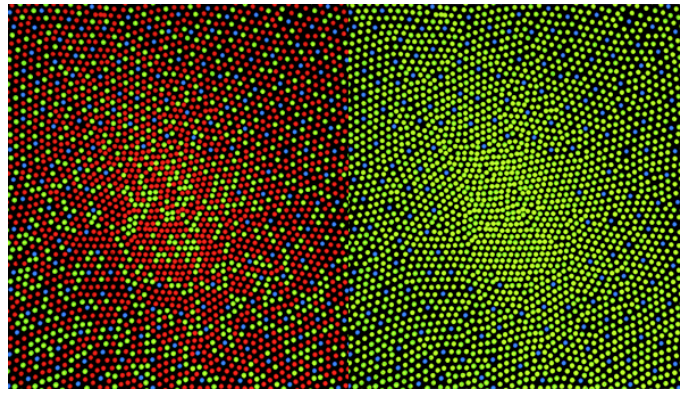
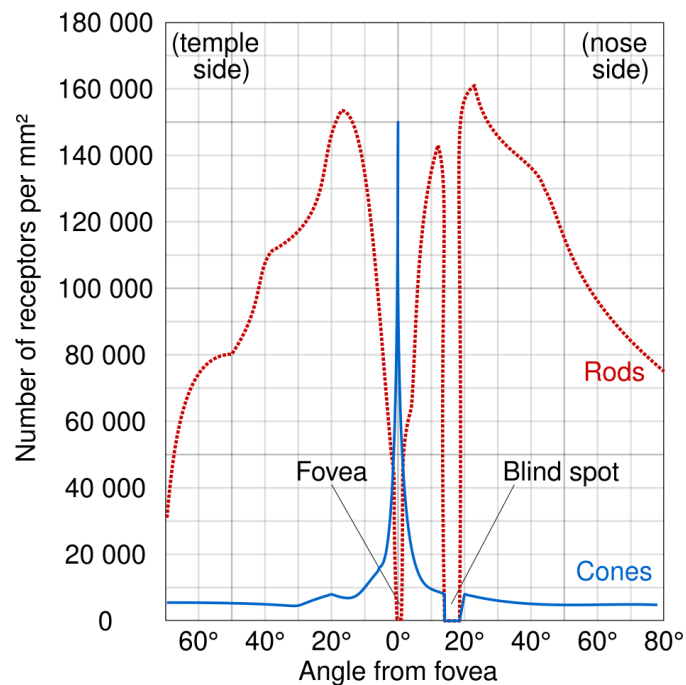


Illustration of the distribution of cone cells in the fovea of an individual with normal colour vision (left), and a colourblind (protanopic) retina. The center of the fovea holds very few blue-sensitive cones.



Distribution of rods and cones along a line passing through the fovea and the blind spot of a human eye^[14]

The vertebrate retina has 10 distinct layers.^[15] From closest to farthest from the vitreous body:

1. **Inner limiting membrane** – basement membrane elaborated by **Müller cells**
2. **Nerve fiber layer** – axons of the **ganglion cell** bodies (a thin layer of Müller cell footplates exists between this layer and the inner limiting membrane)
3. **Ganglion cell layer** – contains nuclei of ganglion cells, the axons of which become the optic nerve fibres, and some displaced **amacrine cells**^[2]
4. **Inner plexiform layer** – contains the synapse between the **bipolar cell** axons and the dendrites of the **ganglion** and amacrine cells^[2]
5. **Inner nuclear layer** – contains the nuclei and surrounding cell bodies (perikarya) of the **amacrine cells**, **bipolar cells**, and **horizontal cells**^[2]

6. **Outer plexiform layer** – projections of rods and cones ending in the rod spherule and cone pedicle, respectively, these make synapses with dendrites of bipolar cells and horizontal cells.^[2] In the **macular** region, this is known as the *fiber layer of Henle*.
7. **Outer nuclear layer** – cell bodies of rods and cones
8. **External limiting membrane** – layer that separates the inner segment portions of the photoreceptors from their cell nuclei
9. **Inner segment / outer segment layer** – inner segments and outer segments of rods and cones, the outer segments contain a highly specialized light-sensing apparatus.^{[16][17]}
10. **Retinal pigment epithelium** – single layer of cuboidal epithelial cells (with extrusions not shown in diagram). This layer is closest to the choroid, and provides nourishment and supportive functions to the neural retina, The black pigment melanin in the pigment layer prevents light reflection throughout the globe of the eyeball; this is extremely important for clear vision.^{[18][19][20]}

These layers can be grouped into four main processing stages—photoreception; transmission to **bipolar cells**; transmission to **ganglion cells**, which also contain photoreceptors, the **photosensitive ganglion cells**; and transmission along the optic nerve. At each synaptic stage, **horizontal** and **amacrine cells** also are laterally connected.

The **optic nerve** is a central tract of many axons of ganglion cells connecting primarily to the **lateral geniculate body**, a visual relay station in the **diencephalon** (the rear of the forebrain). It also projects to the **superior colliculus**, the **suprachiasmatic nucleus**, and the **nucleus of the optic tract**. It passes through the other layers, creating the **optic disc** in primates.^[21]

Additional structures, not directly associated with vision, are found as outgrowths of the retina in some vertebrate groups. In **birds**, the **pecten** is a vascular structure of complex shape that projects from the retina into the **vitreous humour**; it supplies oxygen and nutrients to the eye, and may also aid in vision. **Reptiles** have a similar, but much simpler, structure.^[22]

In adult humans, the entire retina is about 72% of a sphere about 22 mm in diameter. The entire retina contains about 7 million cones and 75 to 150 million rods. The optic disc, a part of the retina sometimes called "the blind spot" because it lacks photoreceptors, is located at the **optic papilla**, where the optic-nerve fibres leave the eye. It appears as an oval white area of 3 mm². Temporal (in the direction of the temples) to this disc is the **macula**, at whose centre is the **fovea**, a pit that is responsible for sharp central vision, but is actually less sensitive to light because of its lack of rods. Human and non-human **primates** possess one fovea, as opposed to certain bird species, such as hawks, that are bifoviate, and dogs and cats, that possess no fovea, but a central band known as the visual streak.^[23] Around the fovea extends the central retina for about 6 mm and then the peripheral retina. The farthest edge of the retina is defined by the **ora serrata**.

The distance from one ora to the other (or macula), the most sensitive area along the [horizontal meridian](#), is about 32 mm.^[24]

In section, the retina is no more than 0.5 mm thick. It has three layers of [nerve](#) cells and two of [synapses](#), including the unique [ribbon synapse](#). The optic nerve carries the [ganglion-cell axons](#) to the brain, and the blood vessels that supply the retina. The ganglion cells lie innermost in the eye while the photoreceptive cells lie beyond. Because of this counter-intuitive arrangement, light must first pass through and around the ganglion cells and through the thickness of the retina, (including its capillary vessels, not shown) before reaching the rods and cones. Light is absorbed by the [retinal pigment epithelium](#) or the [choroid](#) (both of which are opaque).

The [white blood cells](#) in the [capillaries](#) in front of the photoreceptors can be perceived as tiny bright moving dots when looking into blue light. This is known as the [blue field entoptic phenomenon](#) (or Scheerer's phenomenon).

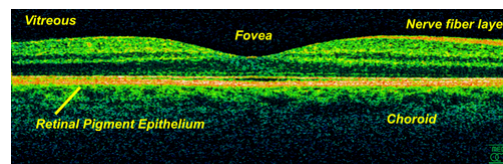
Between the [ganglion-cell](#) layer and the rods and cones are two layers of [neuropils](#), where synaptic contacts are made. The neuropil layers are the [outer plexiform layer](#) and the [inner plexiform layer](#). In the outer neuropil layer, the rods and cones connect to the vertically running [bipolar cells](#), and the horizontally oriented [horizontal cells](#) connect to ganglion cells.

The central retina predominantly contains cones, while the peripheral retina predominantly contains rods. In total, the retina has about seven million cones and a hundred million rods. At the centre of the macula is the foveal pit where the cones are narrow and long, and arranged in a hexagonal [mosaic](#), the most dense, in contradistinction to the much fatter cones located more peripherally in the retina.^[25] At the foveal pit, the other retinal layers are displaced, before building up along the foveal slope until the rim of the fovea, or [parafovea](#), is reached, which is the thickest portion of the retina. The macula has a yellow pigmentation, from screening pigments, and is known as the macula lutea. The area directly surrounding the fovea has the highest density of rods converging on single bipolar cells. Since its cones have a much lesser convergence of signals, the fovea allows for the sharpest vision the eye can attain.^[2]

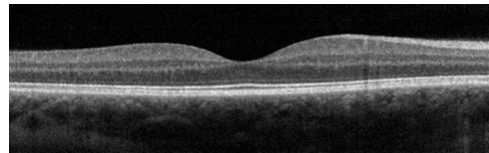
Though the rod and cones are a [mosaic](#) of sorts, transmission from receptors, to bipolars, to [ganglion cells](#) is not direct. Since about 150 million receptors and only 1 million optic nerve fibres exist, convergence and thus mixing of signals must occur. Moreover, the horizontal action of the [horizontal](#) and [amacrine cells](#) can allow one area of the retina to control another (e.g. one stimulus inhibiting another). This inhibition is key to lessening the sum of messages sent to the higher regions of the brain. In some lower vertebrates (e.g. the [pigeon](#)), control of messages is "centrifugal" – that is, one layer can control another, or higher regions of the brain can drive the retinal nerve cells, but in primates, this does not occur.^[2]

Layers imagable with optical coherence tomography

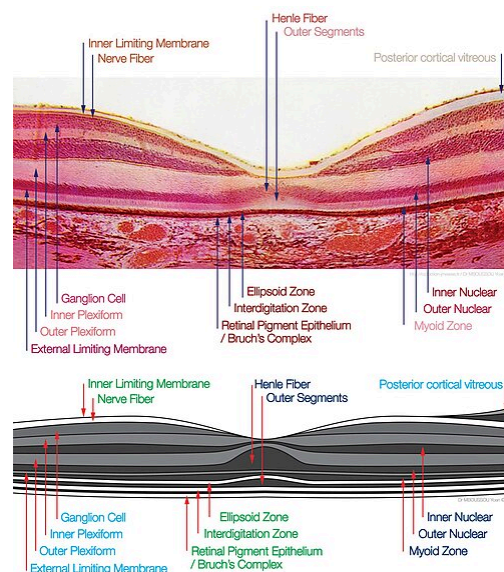
Using [optical coherence tomography](#) (OCT), at least 13 layers can be identified in the retina. The layers and anatomical correlation are:^{[26][27][28]}



Time-Domain OCT of the macular area of a retina at 800 nm, axial resolution 3 μm



Spectral-Domain OCT macula cross-section scan



Macula histology (OCT)

From innermost to outermost, the layers identifiable by OCT are as follows:

#	OCT layer / Conventional label	Anatomical correlate	Reflectivity on OCT	Specific anatomical boundaries?	Ref.
1	Posterior cortical vitreous	Posterior cortical vitreous	Hyper- reflective	Yes	[27]
2	Preretinal space	<i>In eyes where the vitreous has fully or partially detached from the retina, this is the space created between the posterior cortical vitreous face and the internal limiting membrane of the retina.</i>	Hypo- reflective		[27]
3	Internal limiting membrane (ILM)	Formed by Müller cell endfeet (unclear if it can be observed on OCT)	Hyper- reflective	No	[27]
	Nerve fiber layer (NFL)	Ganglion cell axons travelling towards the optic nerve			
4	Ganglion cell layer (GCL)	Ganglion cell bodies (and some displaced amacrine cells)	Hypo- reflective		[27]
5	Inner plexiform layer (IPL)	Synapses between bipolar, amacrine and ganglion cells	Hyper- reflective		[27]
6	Inner nuclear layer (INL)	a) Horizontal, bipolar and amacrine cell bodies b) Müller cell nuclei	Hypo- reflective		[27]
7	Outer plexiform layer (OPL)	Synapses between photoreceptor, bipolar and horizontal cells	Hyper- reflective		[27]
8	(Inner half) Henle's nerve fiber layer (HL)	Photoreceptor axons (obliquely orientated fibres; not present in mid-peripheral or peripheral retina)	Hypo- reflective	No	[27]
	(Outer half) Outer nuclear layer (ONL)	The photoreceptor cell bodies			
9	External limiting membrane (ELM)	Made of zonulae adherens between Müller cells and photoreceptor inner segments	Hyper- reflective		[27]
10	Myoid zone (MZ)	The innermost portion of the photoreceptor inner segment (IS) containing:	Hypo- reflective	No	[29][30]

		• Smooth and rough endoplasmic reticulum • Ribosomes • Golgi bodies • Microtubules • (rarely: <i>mitochondria</i>)			
11	Ellipsoid zone (EZ) IS/OS junction or Photoreceptor integrity line (PIL)	The outermost portion of the photoreceptor inner segment (IS) packed with mitochondria The photoreceptor connecting cilia which bridge the inner and outer segments of the photoreceptor cells.	Very Hyper-reflective	No	[26][31][29][27][32][33]
12	Photoreceptor outer segments (OS)	The photoreceptor outer segments (OS) which contain disks filled with opsin , the molecule that absorbs photons.	Hypo-reflective		
13	Interdigitation zone (IZ)	Apices of the RPE cells which encase part of the cone OSs. <i>Poorly distinguishable from RPE. Previously: "cone outer segment tips line" (COST)</i>	Hyper-reflective	No	[34][27]
14	RPE/Bruch's complex	RPE phagosome zone RPE melanosome zone RPE mitochondria zone + Junction between the RPE & Bruch's membrane	Very Hyper-reflective Hypo-reflective Very Hyper-reflective	No	[26][27]
15	Choriocapillaris	Thin layer of moderate reflectivity in inner choroid		No	[27]
16	Sattler's layer	Thick layer of round or ovalshaped hyperreflective profiles, with hyporefective cores in mid-choroid			[27]
17	Haller's layer	Thick layer of oval-shaped hyperreflective profiles, with hyporefective cores in outer choroid			[27]
18	Choroidal-scleral juncture	Zone at the outer choroid with a marked change in texture, in which large circular or ovoid profiles abut a homogenous region of variable reflectivity			[27]

Development

Retinal development begins with the establishment of the eye fields mediated by the **SHH** and **SIX3** proteins, with subsequent development of the optic vesicles regulated by the **PAX6** and

LHX2 proteins.^[35] The role of Pax6 in eye development was elegantly demonstrated by Walter Gehring and colleagues, who showed that ectopic expression of Pax6 can lead to eye formation on *Drosophila* antennae, wings, and legs.^[36] The optic vesicle gives rise to three structures: the neural retina, the retinal pigmented epithelium, and the optic stalk. The neural retina contains the retinal progenitor cells (RPCs) that give rise to the seven cell types of the retina. Differentiation begins with the **retinal ganglion cells** and concludes with production of the Muller glia.^[37] Although each cell type differentiates from the RPCs in a sequential order, there is considerable overlap in the timing of when individual cell types differentiate.^[35] The cues that determine a RPC daughter cell fate are coded by multiple transcription factor families including the **bHLH** and **homeodomain** factors.^{[38][39]}

In addition to guiding cell fate determination, cues exist in the retina to determine the dorsal-ventral (D-V) and nasal-temporal (N-T) axes. The D-V axis is established by a ventral to dorsal gradient of **VAX2**, whereas the N-T axis is coordinated by expression of the forkhead transcription factors **FOXD1** and **FOXG1**. Additional gradients are formed within the retina.^[39] This spatial distribution may aid in proper targeting of RGC axons that function to establish the retinotopic map.^[35]

Blood supply



Fundus photograph showing the blood vessels in a normal human retina. Veins are darker and slightly wider than corresponding arteries. The **optic disc** is at right, and the **macula lutea** is near the centre.

The retina is stratified into distinct layers, each containing specific cell types or cellular compartments^[40] that have metabolisms with different nutritional requirements.^[41] To satisfy these requirements, the **ophthalmic artery** bifurcates and supplies the retina via two distinct vascular networks: the choroidal network, which supplies the choroid and the outer retina, and the retinal network, which supplies the retina's inner layer.^[42]

Although the inverted retina of vertebrates appears counter-intuitive, it is necessary for the proper functioning of the retina. The photoreceptor layer must be embedded in the retinal pigment epithelium (RPE), which performs at least seven vital functions,^[43] one of the most

obvious being to supply oxygen and other necessary nutrients needed for the photoreceptors to function.

Energy requirements

The energy requirements of the retina are even greater than that of the brain.^[4] This is due to the additional energy needed to continuously renew the photoreceptor outer segments, of which 10% are shed daily.^[4] Energy demands are greatest during dark adaptation when its sensitivity is most enhanced.^[44] The choroid supplies about 75% of these nutrients to the retina and the retinal vasculature only 25%.^[5]

When light strikes 11-cis-retinal (in the disks in the rods and cones), 11-cis-retinal changes to all-trans-retinal which then triggers changes in the opsins. Now, the outer segments do not regenerate the retinal back into the cis- form once it has been changed by light. Instead the retinal is pumped out to the surrounding RPE where it is regenerated and transported back into the outer segments of the photoreceptors. This recycling function of the RPE protects the photoreceptors against photo-oxidative damage^{[45][46]} and allows the photoreceptor cells to have decades-long useful lives.

In birds

The bird retina is devoid of blood vessels, perhaps to give unobscured passage of light for forming images, thus giving better resolution. It is, therefore, a considered view that the bird retina depends for nutrition and oxygen supply on a specialized organ, called the "pecten" or [pecten oculi](#), located on the blind spot or optic disk. This organ is extremely rich in blood vessels and is thought to supply nutrition and oxygen to the bird retina by diffusion through the vitreous body. The pecten is highly rich in alkaline phosphatase activity and polarized cells in its bridge portion – both befitting its secretory role.^[47] Pecten cells are packed with dark melanin granules, which have been theorized to keep this organ warm with the absorption of stray light falling on the pecten. This is considered to enhance metabolic rate of the pecten, thereby exporting more nutritive molecules to meet the stringent energy requirements of the retina during long periods of exposure to light.^[48]

Biometric identification and diagnosis of disease

The bifurcations and other physical characteristics of the inner retinal vascular network are known to vary among individuals,^[49] and these individual variances have been used for [biometric identification](#) and for early detection of the onset of disease. The mapping of vascular bifurcations is one of the basic steps in biometric identification.^[50] Results of such analyses of retinal blood vessel structure can be evaluated against the ground truth data^[51] of vascular bifurcations of retinal fundus images that are obtained from the DRIVE dataset.^[52] In addition, the classes of vessels of the DRIVE dataset have also been identified,^[53] and an automated method for accurate extraction of these bifurcations is also available.^[54] Changes in retinal blood

circulation are seen with aging^[55] and exposure to air pollution,^[56] and may indicate cardiovascular diseases such as hypertension and atherosclerosis.^{[57][58][59]} Determining the equivalent width of arterioles and venules near the optic disc is also a widely used technique to identify cardiovascular risks.^[60]

Функция

Сетчатка преобразует оптическое изображение в нервные импульсы, начиная с упорядоченного возбуждения светочувствительных пигментов палочек и колбочек сетчатки — **фоторецепторных клеток**. Возбуждение обрабатывается нервной системой и различными отделами мозга, которые работают параллельно, формируя в мозге представление о внешней среде.

The cones respond to bright light and mediate high-resolution colour vision during daylight illumination (also called **photopic vision**). The rod responses are saturated at daylight levels and do not contribute to pattern vision. However, rods do respond to dim light and mediate lower-resolution, monochromatic vision under very low levels of illumination (called **scotopic vision**). The illumination in most office settings falls between these two levels and is called **mesopic vision**. At mesopic light levels, both the rods and cones are actively contributing pattern information. What contribution the rod information makes to pattern vision under these circumstances is unclear.

The response of cones to various wavelengths of light is called their spectral sensitivity. In normal human vision, the spectral sensitivity of a cone falls into one of three subtypes, often called blue, green, and red, but more accurately known as short, medium, and long wavelength-sensitive cone subtypes. It is a lack of one or more of the cone subtypes that causes individuals to have deficiencies in colour vision or various kinds of **colour blindness**. These individuals are not blind to objects of a particular colour, but are unable to distinguish between colours that can be distinguished by people with normal vision. Humans have this **trichromatic vision**, while most other mammals lack cones with red sensitive pigment and therefore have poorer dichromatic colour vision. However, some animals have four spectral subtypes, e.g. the trout adds an ultraviolet subgroup to short, medium, and long subtypes that are similar to humans. Some fish are sensitive to the polarization of light as well.

In the photoreceptors, exposure to light hyperpolarizes the membrane in a series of graded shifts. The outer cell segment contains a **photopigment**. Inside the cell the normal levels of **cyclic guanosine monophosphate** (cGMP) keep the Na⁺ channel open, and thus in the resting state the cell is depolarised. The **photon** causes the **retinal** bound to the receptor protein to **isomerise** to **trans-retinal**. This causes the receptor to activate multiple **G-proteins**. This in turn causes the G_s subunit of the protein to activate a phosphodiesterase (PDE6), which degrades cGMP, resulting in

the closing of Na⁺ [cyclic nucleotide-gated ion channels](#) (CNGs). Thus the cell is hyperpolarised. The amount of neurotransmitter released is reduced in bright light and increases as light levels fall. The actual photopigment is bleached away in bright light and only replaced as a chemical process, so in a transition from bright light to darkness the eye can take up to thirty minutes to reach full sensitivity.

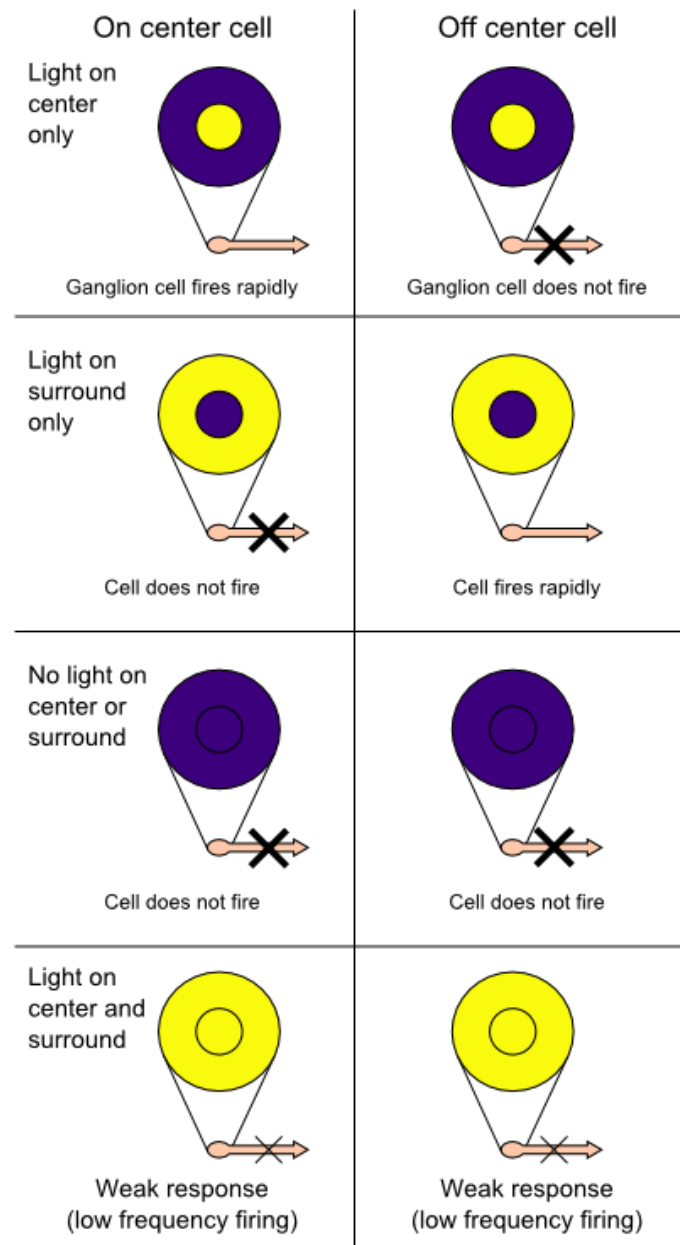
When thus excited by light, the photoreceptor sends a proportional response [synaptically](#) to [bipolar cells](#) which in turn signal the [retinal ganglion cells](#). The photoreceptors are also cross-linked by [horizontal cells](#) and [amacrine cells](#), which modify the synaptic signal before it reaches the ganglion cells, the neural signals being intermixed and combined. Of the retina's nerve cells, only the retinal ganglion cells and few amacrine cells create [action potentials](#).

In the retinal ganglion cells there are two types of response, depending on the [receptive field](#) of the cell. The receptive fields of retinal ganglion cells comprise a central, approximately circular area, where light has one effect on the firing of the cell, and an annular surround, where light has the opposite effect. In ON cells, an increment in light intensity in the centre of the receptive field causes the firing rate to increase. In OFF cells, it makes it decrease. In a linear model, this response profile is well described by a [difference of Gaussians](#) and is the basis for [edge detection](#) algorithms. Beyond this simple difference, ganglion cells are also differentiated by chromatic sensitivity and the type of spatial summation. Cells showing linear spatial summation are termed X cells (also called parvocellular, P, or midget ganglion cells), and those showing non-linear summation are Y cells (also called magnocellular, M, or parasol retinal ganglion cells), although the correspondence between X and Y cells (in the cat retina) and P and M cells (in the primate retina) is not as simple as it once seemed.

In the transfer of visual signals to the brain, the [visual pathway](#), the retina is vertically divided in two, a temporal (nearer to the temple) half and a nasal (nearer to the nose) half. The axons from the nasal half cross the brain at the [optic chiasma](#) to join with axons from the temporal half of the other eye before passing into the [lateral geniculate body](#).

Although there are more than 130 million retinal receptors, there are only approximately 1.2 million fibres (axons) in the optic nerve. So, a large amount of pre-processing is performed within the retina. The [fovea](#) produces the most accurate information. Despite occupying about 0.01% of the visual field (less than 2° of [visual angle](#)), about 10% of axons in the optic nerve are devoted to the fovea. The resolution limit of the fovea has been determined to be around 10,000 points. The information capacity is estimated at 500,000 bits per second (for more information on bits, see [information theory](#)) without colour or around 600,000 bits per second including colour.^[61]

Spatial encoding



On-centres and off-centres of the retina

When the retina sends neural impulses representing an image to the brain, it spatially encodes (compresses) those impulses to fit the limited capacity of the optic nerve. Compression is necessary because there are 100 times more **photoreceptor cells** than **ganglion cells**. This is done by "**decorrelation**", which is carried out by the "centre-surround structures", which are implemented by the bipolar and ganglion cells.

There are two types of centre-surround structures in the retina – on-centres and off-centres. On-centres have a positively weighted centre and a negatively weighted surround. Off-centres are just the opposite. Positive weighting is more commonly known as **excitatory**, and negative weighting as **inhibitory**.

These centre-surround structures are not physical apparent, in the sense that one cannot see them by staining samples of tissue and examining the retina's anatomy. The centre-surround structures are logical (i.e., mathematically abstract) in the sense that they depend on the

connection strengths between bipolar and ganglion cells. It is believed that the connection strength between cells is caused by the number and types of [ion channels](#) embedded in the [synapses](#) between the bipolar and ganglion cells.

The centre-surround structures are mathematically equivalent to the [edge detection](#) algorithms used by computer programmers to extract or enhance the edges in a digital photograph. Thus, the retina performs operations on the image-representing impulses to enhance the edges of objects within its visual field. For example, in a picture of a dog, a cat and a car, it is the edges of these objects that contain the most information. In order for higher functions in the brain (or in a computer for that matter) to extract and classify objects such as a dog and a cat, the retina is the first step to separating out the various objects within the scene.

As an example, the following [matrix](#) is at the heart of a computer [algorithm](#) that implements edge detection. This matrix is the computer equivalent to the centre-surround structure. In this example, each box (element) within this matrix would be connected to one photoreceptor. The photoreceptor in the centre is the current receptor being processed. The centre photoreceptor is multiplied by the +1 weight factor. The surrounding photoreceptors are the "nearest neighbors" to the centre and are multiplied by the $-1/8$ value. The sum of all nine of these elements is finally calculated. This summation is repeated for every photoreceptor in the image by shifting left to the end of a row and then down to the next line.

$-1/8$	$-1/8$	$-1/8$
$-1/8$	+1	$-1/8$
$-1/8$	$-1/8$	$-1/8$

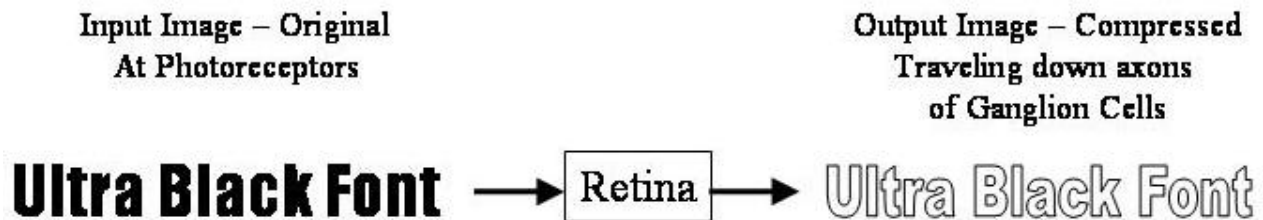
The total sum of this matrix is zero, if all the inputs from the nine photoreceptors are of the same value. The zero result indicates the image was uniform (non-changing) within this small patch. Negative or positive sums mean the image was varying (changing) within this small patch of nine photoreceptors.

The above matrix is only an approximation to what really happens inside the retina. The differences are:

- The above example is called "balanced". The term balanced means that the sum of the negative weights is equal to the sum of the positive weights so that they cancel out perfectly. Retinal ganglion cells are almost never perfectly balanced.
- The table is square while the centre-surround structures in the retina are circular.
- Neurons operate on [spike trains](#) traveling down nerve cell [axons](#). Computers operate on a single [floating-point number](#) that is essentially constant from each input [pixel](#). (The computer pixel is basically the equivalent of a biological photoreceptor.)

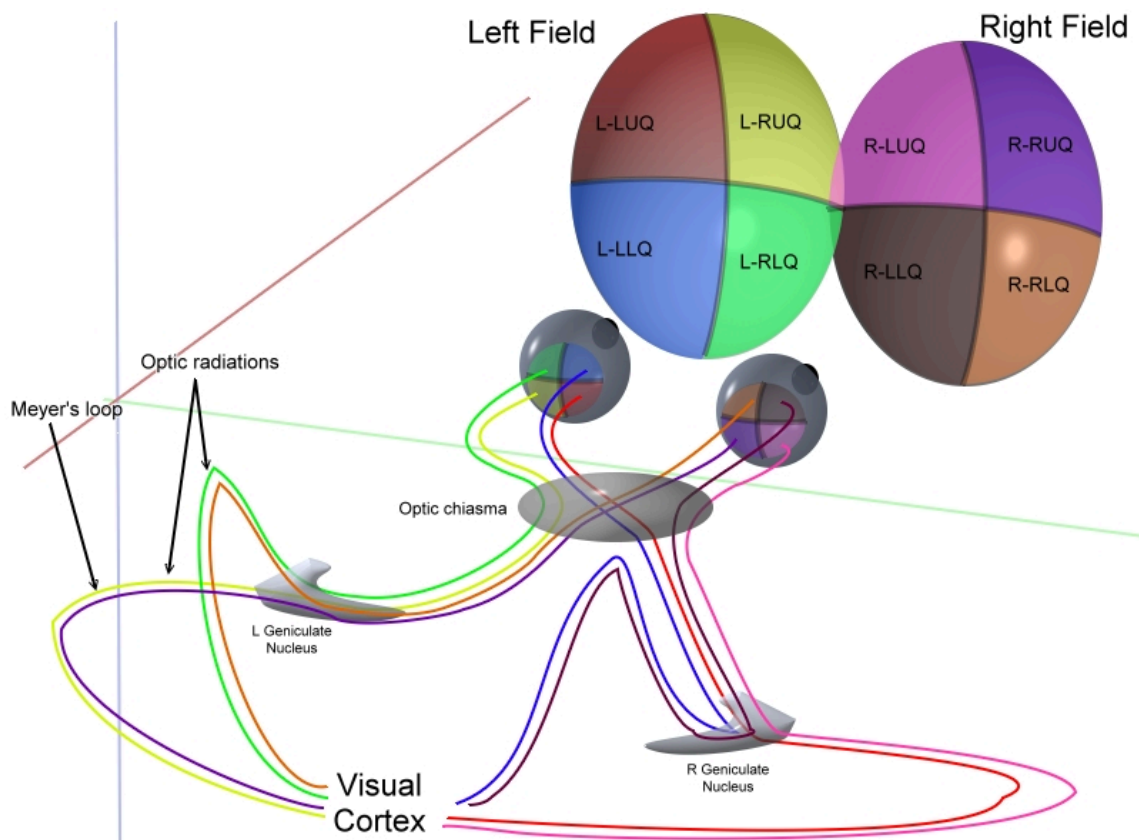
- The retina performs all these calculations in parallel while the computer operates on each pixel one at a time. The retina performs no repeated summations and shifting as would a computer.
- Finally, the [horizontal](#) and [amacrine cells](#) play a significant role in this process, but that is not represented here.

Here is an example of an input image and how edge detection would modify it.



Once the image is spatially encoded by the centre-surround structures, the signal is sent out along the optic nerve (via the axons of the ganglion cells) through the [optic chiasm](#) to the LGN ([lateral geniculate nucleus](#)). The exact function of the LGN is unknown at this time. The output of the LGN is then sent to the back of the brain. Specifically, the output of the LGN "radiates" out to the V1 [primary visual cortex](#).

Simplified signal flow: Photoreceptors → Bipolar → Ganglion → Chiasm → LGN → V1 cortex



Клиническая значимость

There are many inherited and acquired diseases or disorders that may affect the retina. Some of them include:

- **Пигментная дистрофия сетчатки** — это группа генетических заболеваний, поражающих сетчатку глаза и приводящих к потере ночного и периферического зрения.
- **Дегенерация макулы** — это группа заболеваний, характеризующихся потерей центрального зрения из-за гибели или повреждения клеток в **макуле**.
- **Дистрофия колбочек и палочек (ДКП)** — это группа заболеваний, при которых потеря зрения вызвана разрушением **колбочек** и/или **палочек** в сетчатке глаза.
- При **отслоении сетчатки** сетчатка отделяется от задней поверхности глазного яблока. **Игнипунктура** — устаревший метод лечения. Термин **отслоение сетчатки** используется для описания отделения нейросенсорной сетчатки от **пигментного эпителия сетчатки**.^[62] Существует несколько современных методов лечения отслойки сетчатки: **пневматическая ретинопексия**, **склеральная подтяжка**, **криотерапия**, **лазерная фотокоагуляция** и **витрэктомия** через плоскую часть цилиарного тела.
- Как **гипертония**, так и **сахарный диабет** могут привести к повреждению крошечных кровеносных сосудов, питающих сетчатку, что приводит к **гипертонической ретинопатии** и **диабетической ретинопатии**.
- **Ретинобластома** — это **рак** сетчатки.
- К заболеваниям сетчатки у **собак** относятся **дисплазия сетчатки**, **прогрессирующая атрофия сетчатки** и **внезапная приобретенная дегенерация сетчатки**.
- ***Lipaemia retinalis*** is a white appearance of the retina, and can occur by lipid deposition in **lipoprotein lipase deficiency**.
- ***Retinal Detachment***. The neural retina occasionally detaches from the pigment epithelium. In some instances, the cause of such detachment is injury to the eyeball that allows fluid or blood to collect between the neural retina and the pigment epithelium. Detachment is occasionally caused by contracture of fine collagenous fibrils in the vitreous humor, which pull areas of the retina toward the interior of the globe.^[25]
- ***Night blindness***: Night blindness occurs in any person with severe vitamin A deficiency. The reason for this is that without vitamin A, the amounts of retinal and rhodopsin that can be formed are severely depressed. This condition is called night blindness because the amount of light available at night is too little to permit adequate vision in vitamin A–deficient persons.^[18]

In addition, the retina has been described as a "window" into the brain and body, given that abnormalities detected through an examination of the retina can discover both neurological and

systemic diseases.^[63]

Diagnosis

A number of different instruments are available for the diagnosis of diseases and disorders affecting the retina. [Ophthalmoscopy](#) and [fundus photography](#) have long been used to examine the retina. Recently, [adaptive optics](#) has been used to image individual rods and cones in the living human retina, and a company based in Scotland has engineered technology that allows physicians to observe the complete retina without any discomfort to patients.^[64]

The [electroretinogram](#) is used to [non-invasively](#) measure the retina's electrical activity, which is affected by certain diseases. A relatively new technology, now becoming widely available, is [optical coherence tomography](#) (OCT). This non-invasive technique allows one to obtain a [3D](#) volumetric or high resolution cross-sectional [tomogram](#) of the fine structures of the retina, with [histologic](#) quality. [Retinal vessel analysis](#) is a non-invasive method to examine the small arteries and veins in the retina which allows to draw conclusions about the morphology and the function of small vessels elsewhere in the human body. It has been established as a predictor of [cardiovascular disease](#)^[65] and seems to have, according to a study published in 2019, potential in the early detection of Alzheimer's disease.^[66]

Treatment

Treatment depends upon the nature of the disease or disorder.

Common treatment modalities

The following are commonly modalities of management for retinal disease:

- [Intravitreal](#) medication, such as [anti-VEGF](#) or [corticosteroid](#) agents
- [Vitreoretinal surgery](#)
- Use of nutritional supplements
- Modification of systemic risk factors for retinal disease

Uncommon treatment modalities

Retinal gene therapy

[Gene therapy](#) holds promise as a potential avenue to cure a wide range of retinal diseases. This involves using a non-infectious virus to shuttle a gene into a part of the retina. Recombinant [adeno-associated virus](#) (rAAV) vectors possess a number of features that render them ideally suited for retinal gene therapy, including a lack of pathogenicity, minimal immunogenicity, and the ability to transduce postmitotic cells in a stable and efficient manner.^[67] rAAV vectors are increasingly utilized for their ability to mediate efficient transduction of [retinal pigment epithelium](#) (RPE), [photoreceptor cells](#) and [retinal ganglion cells](#). Each cell type can be specifically

targeted by choosing the appropriate combination of AAV [serotype](#), promoter, and intraocular injection site.

Several clinical trials have already reported positive results using rAAV to treat [Leber's congenital amaurosis](#), showing that the therapy was both safe and effective.^{[68][69]} There were no serious adverse events, and patients in all three studies showed improvement in their visual function as measured by a number of methods. The methods used varied among the three trials, but included both functional methods such as visual acuity^{[69][70][71]} and functional mobility^{[70][71][72]} as well as objective measures that are less susceptible to bias, such as the pupil's ability to respond to light^{[68][73]} and improvements on functional MRI.^[74] Improvements were sustained over the long-term, with patients continuing to do well after more than 1.5 years.^{[68][69]}

The unique architecture of the retina and its relatively immune-privileged environment help this process.^[75] [Tight junctions](#) that form the [blood retinal barrier](#) separate the subretinal space from the blood supply, thus protecting it from microbes and most immune-mediated damage, and enhancing its potential to respond to vector-mediated therapies. The highly compartmentalized anatomy of the eye facilitates accurate delivery of therapeutic vector suspensions to specific tissues under direct visualization using microsurgical techniques.^[76] In the sheltered environment of the retina, AAV vectors are able to maintain high levels of [transgene](#) expression in the retinal pigmented epithelium (RPE), photoreceptors, or ganglion cells for long periods of time after a single treatment. In addition, the eye and the visual system can be routinely and easily monitored for visual function and retinal structural changes after injections with noninvasive advanced technology, such as visual acuities, [contrast sensitivity](#), [fundus](#) auto-fluorescence (FAF), dark-adapted visual thresholds, vascular diameters, pupillometry, [electroretinography](#) (ERG), multifocal ERG and [optical coherence tomography](#) (OCT).^[77]

This strategy is effective against a number of retinal diseases that have been studied, including neovascular diseases that are features of [age-related macular degeneration](#), [diabetic retinopathy](#) and [retinopathy of prematurity](#). Since the regulation of vascularization in the mature retina involves a balance between endogenous positive [growth factors](#), such as [vascular endothelial growth factor](#) (VEGF) and inhibitors of [angiogenesis](#), such as pigment epithelium-derived factor (PEDF), rAAV-mediated expression of PEDF, angiostatin, and the soluble VEGF receptor sFlt-1, which are all antiangiogenic proteins, have been shown to reduce aberrant vessel formation in animal models.^[78] Since specific gene therapies cannot readily be used to treat a significant fraction of patients with retinal dystrophy, there is a major interest in developing a more generally applicable survival factor therapy. [Neurotrophic factors](#) have the ability to modulate neuronal growth during development to maintain existing cells and to allow recovery of injured neuronal populations in the eye. AAV encoding neurotrophic factors such as fibroblast growth factor (FGF) family members and GDNF either protected photoreceptors from apoptosis or slowed down cell death.^[78]

Organ transplantation In 2023, [transplantation](#) of face and an eye was attempted, with limited success.^[79] Whole [eye transplantation](#) is fraught with complications such as viability of the donor eye, regeneration of neurons servicing the eye, immunological acceptance and ethical issues.^[80] At [MIT](#), The University of Southern California, RWTH Aachen University, and the [University of New South Wales](#), an "artificial retina" is under development: an implant which will bypass the photoreceptors of the retina and stimulate the attached nerve cells directly, with signals from a digital camera.

Electronic assistance

In 2017, a patient was implanted with an electrode implant designed to assist patients with retinitis pigmentosa to detect light.^[81] In 2025, a small camera mounted on glasses and a wireless chip inserted into the eye of patients with macular degeneration restored functional vision (such as shapes and forms rather than simply light detection).^[82] After completion of a one year trial, 27 of 32 patients could read at least two additional lines on the standard eye chart, but importantly could read subway signs and food labels.

История

Около 300 года [до н. э. Герофил](#) описал сетчатку, препарирова глаза трупов. Он назвал ее *паутинным* слоем из-за сходства с паутиной и *сетчатым* слоем из-за сходства с рыболовной сетью. Термин [паутинный](#) стал обозначать слой вокруг головного мозга, а термин *сетчатый* — *сетчатку*.^[83]

Between 1011 and 1021 CE, [Ibn Al-Haytham](#) published numerous experiments demonstrating that sight occurs from light reflecting from objects into the eye. This is consistent with [intromission theory](#) and against [emission theory](#), the theory that sight occurs from rays emitted by the eyes. However, Ibn Al-Haytham decided that the retina could not be responsible for the beginnings of vision because the image formed on it was inverted. Instead he decided it must begin at the surface of the lens.^[84]

In 1604, [Johannes Kepler](#) worked out the optics of the eye and decided that the retina must be where sight begins. He left it up to other scientists to reconcile the inverted retinal image with our perception of the world as upright.^[85]

In 1894, [Santiago Ramón y Cajal](#) published the first major characterization of retinal neurons in *Retina der Wirbelthiere* (*The Retina of Vertebrates*).^[86]

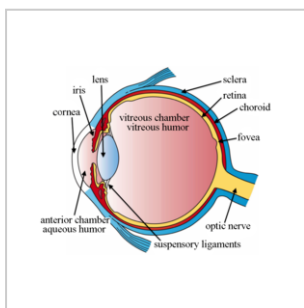
[George Wald](#), [Haldan Keffer Hartline](#), and [Ragnar Granit](#) won the 1967 [Nobel Prize in Physiology or Medicine](#) for their scientific research on the retina.^[87]

A recent [University of Pennsylvania](#) study calculated that the approximate [bandwidth](#) of human retinae is 8.75 megabits per second, whereas a [guinea pig](#)'s retinal transfer rate is 875 kilobits per second.^[88]

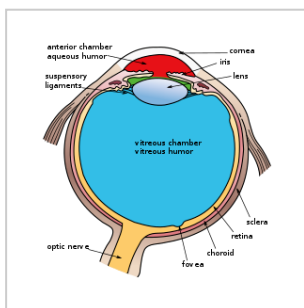
[MacLaren](#) & Pearson and colleagues at [University College London](#) and [Moorfields Eye Hospital](#) in London, in 2006, showed that [photoreceptor cells](#) could be transplanted successfully in the mouse retina if donor cells were at a critical developmental stage.^[89] Recently Ader and colleagues in Dublin showed, using the electron microscope, that transplanted photoreceptors formed synaptic connections.^[90]

In 2012, [Sebastian Seung](#) and his laboratory at [MIT](#) launched [EyeWire](#), an online [Citizen science](#) game where players trace neurons in the retina.^[91] The goals of the EyeWire project are to identify specific cell types within the known broad classes of retinal cells, and to [map the connections](#) between neurons in the retina, which will help to determine how vision works.^{[92][93]}

Дополнительные изображения



The structures of the eye labeled



Another view of the eye and the structures of the eye labeled

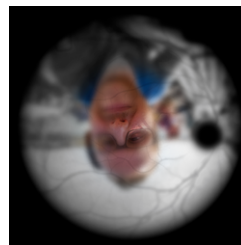


Illustration of image as 'seen' by the retina independent of optic nerve and striate cortex processing

См. также



- [Adeno associated virus and gene therapy of the human retina](#)
- [Charles Schepens](#) – "the father of modern retinal surgery"
- [Evolution of the eye](#)
- [Duplex retina](#)
- [List of xanthoma variants associated with hyperlipoproteinemia subtypes](#)

- [Persistence of vision](#)
- [Retinal scan](#)
- [Retinal vein occlusion](#)
- [Rhodopsin](#)

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External links

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- [Histology of the Eye \(https://www.youtube.com/watch?v=nELWup_Gghw\)](https://www.youtube.com/watch?v=nELWup_Gghw) , edited by William Krause, Dept. Pathology and Anatomical science, University of Missouri School of Medicine
 - [Eye, Brain, and Vision – online book – by David Hubel \(https://web.archive.org/web/20070314135325/http://neuro.med.harvard.edu/site/dh/index.html\)](https://web.archive.org/web/20070314135325/http://neuro.med.harvard.edu/site/dh/index.html)
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 - [Jeremy Nathans's Seminars: "The Vertebrate Retina: Structure, Function, and Evolution" \(http://www.ibiology.org/neuroscience/color-vision/\)](http://www.ibiology.org/neuroscience/color-vision/)
 - [Retina – Cell Centered Database \(http://ccdb.ucsd.edu/sand/main?mpid=54&event=displaySum\)](http://ccdb.ucsd.edu/sand/main?mpid=54&event=displaySum)
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